

Hallucination Is Nothing You Need: Preemptive Detection of VLM Hallucinations

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1 Problem Statement and Background

Vision-language models (VLMs) have become increasingly capable at image captioning, visual question answering, and multimodal reasoning. However, they still suffer from hallucination: the model may describe objects, attributes, counts, relations, or actions that are not actually supported by the image. These errors are especially concerning because VLMs often produce fluent and confident responses even when the visual grounding is wrong.

Most existing hallucination mitigation methods treat the model as a black box. They usually detect hallucinations after generation using an external verifier, another language model, or a post-processing correction module, then ask the model to revise or resample its answer. While useful, this reactive setting misses an opportunity to study the model’s internal mechanism. If hallucination risk is already visible in the hidden states before decoding begins, then we may be able to detect the failure early.

Our project asks: *Can we use internal representations of a VLM to preemptively detect visual hallucination?* This project follows both Theme 1 and Theme 3. It is an evaluation project because we benchmark hallucination detection on VLM datasets. It is also an interpretability project because we analyze which hidden states and layers reveal hallucination risk.

2 Motivation

Our motivation is that hallucination is not only a surface-level decoding problem. It may reflect an internal mismatch between the model’s language-side knowledge and the visual evidence in the input image. When a VLM answers a visual question, the decoder must combine information from visual tokens and text tokens. If the internal states drift toward language priors rather than image-grounded evidence, the model may retrieve

or generate nonexistent information.

Recent work suggests that hidden states can reveal valuable signals about model factuality before the answer is generated. This makes a lightweight internal-state classifier attractive, because it can act as an early warning system for unreliable VLM answers. For a six-week group project, we make detection our main objective and leave steering-based intervention as future work.

3 Related Work

Our work draws on two lines of research: (1) interpreting and probing the internal states of large foundation models for hallucination detection, which is the focus of our study, and (2) steering model representations for more factual or controlled generation, which motivates our future work.

Internal Representations for Hallucination Detection. **InternalInspector** (Beigi et al., 2024) trains a contrastive learning-based classifier on attention states, feed-forward states, and activation states across all layers. It shows that internal states can estimate confidence and identify hallucinated outputs better than logit-based and self-evaluation baselines. However, InternalInspector uses a broad set of internal states and is mainly a confidence estimator after generation, which may be expensive for our setting. We adopt its insight that internal states contain truthfulness information, but we use a smaller feature set focused on selected layers.

More relevant to our project idea goal is **FACTCHECKMATE** (Alnuhait et al., 2025), which demonstrates that a lightweight classifier operating on hidden states produced over the input prompt can preemptively predict whether an LLM will hallucinate before decoding begins. FACTCHECKMATE uses the FFN output at a single mid-to-late layer, mean-pooled over prompt tokens, and achieves strong detection accuracy across multiple QA datasets and model families. This

work provides the main empirical foundation for our detector, but we adapt the idea from text-only LLMs to image-conditioned VLMs.

Steering and Controlling Generation. **DMAS** (Yin et al., 2026) proposes Dynamic Multimodal Activation Steering for VLM hallucination mitigation. It builds truthfulness steering vectors from factual versus hallucinated examples and visual-perception steering vectors from clean versus noisy image inputs, then applies them to selected attention heads. This provides a direct VLM steering inspiration for our future steering goal. **LM-Steers** (Han et al., 2024) shows that simple transformations in output word embedding space can steer language generation attributes with very few parameters. Together, these papers suggest that lightweight steering directions can be effective without full model retraining. All of them can support our future work to steer the model before hallucination result generating.

Positioning Our Work. InternalInspector and FACTCHECKMATE show that hidden states can detect hallucination risk, while DMAS, and LM-Steers show that internal representations can be steered. Our project focuses on the detection side: we train a VLM hallucination detector on pre-decoding hidden states across five open-source models, and analyze which layers and pooling strategies best expose hallucination risk. Our expected contribution is not a new VLM training method, but a reproducible study of whether FFN-based early detection transfers from text-only LLMs to small multimodal models.

4 Approach

4.1 Dataset Construction

We evaluate our approach on the POPE benchmark (Li et al., 2023), a widely used object hallucination evaluation dataset for vision-language models. POPE frames hallucination detection as a binary visual question answering task: given an image, the model is asked whether a specific object is present, and must respond with “yes” or “no.” This formulation makes ground-truth labeling straightforward and deterministic, without requiring an external LLM judge.

For each (image, question) pair in POPE, we run each frozen VLM and collect its generated answer. By comparing the model’s response against the ground-truth label, we assign a binary annota-

tion: **hallucinated** (model answer is incorrect) or **factual** (model answer is correct). This produces a labeled dataset of (hidden state, label) pairs for each model, which is then used to train and evaluate the hallucination classifier.

4.2 Models

We experiment across five open-source VLMs of varying architectures and scales: Qwen2.5-VL-3B, Phi-3.5-Vision-Instruct, InternVL2-2B, DeepSeek-VL-1.3B-Base, and MiniCPM-V2. All models are kept **frozen** throughout all experiments; no VLM parameters are updated at any point. This design choice ensures that our method is purely a lightweight add-on and does not require any re-training of the base model.

4.3 Feature Extraction

Motivated by FactCheckmate (Alnuhait et al., 2025), we hypothesize that the internal hidden states of a VLM, computed over the input prompt before any answer tokens are decoded, already encode signals predictive of whether the model will hallucinate. To test this, we register forward hooks on the feed-forward network (FFN) output at every transformer layer of each model’s language decoder. For a given input, we perform a single forward pass and extract the FFN output at each layer l . We compare two feature aggregation strategies: (i) **mean-pooling** across all prompt token positions, and (ii) the **last-token** representation. Each strategy yields a fixed-size feature vector $\bar{\mathbf{h}}^{(l)}$ per layer, allowing us to study which aggregation better exposes hallucination signal.

This extraction is performed before decoding begins, meaning the classifier operates entirely on pre-generation representations — making detection truly preemptive.

4.4 Hallucination Classifier

For each layer l and each pooling strategy, we train an independent lightweight MLP classifier $f_{\theta}^{(l)}$ on the corresponding extracted feature vectors $\bar{\mathbf{h}}^{(l)}$, with binary cross-entropy loss and binary hallucination labels as supervision. This yields one classifier per layer and methods combination for each model, from which we select the best-performing classifier based on validation F1 score. This layer selection procedure also serves as an interpretability analysis, revealing which layers encode the most hallucination-relevant signal for each VLM architecture.

At inference time, the selected classifier takes the pre-decoding FFN hidden state at the optimal layer as input and outputs a hallucination risk score. The additional compute overhead is minimal: one forward pass for feature extraction plus a single MLP call.

4.5 Future Work: Steering Intervention

A natural extension of our detection framework is a lightweight steering module that, when the classifier flags high hallucination risk, adjusts the VLM’s hidden states before decoding to nudge the model toward a more visually grounded response. We leave this intervention component as future work.

5 Experimental Results

5.1 Main Results

5.1.1 Layer Ablation Results

Qwen2.5-VL-3B-Instruct. Last-token features consistently outperform mean pooling across all layers and peak early at **layer 11** (val acc 0.85, test acc 0.84, test F1 0.37). Mean pooling is more erratic and peaks much later at layer 31 (val acc 0.80, F1 $\approx$ 0.27) [Figure 1], suggesting that averaging over all token positions dilutes the concentrated hallucination signal that the final prompt token carries. This mid-network peak indicates that layer 11 already encodes sufficient multimodal grounding information for binary hallucination prediction.

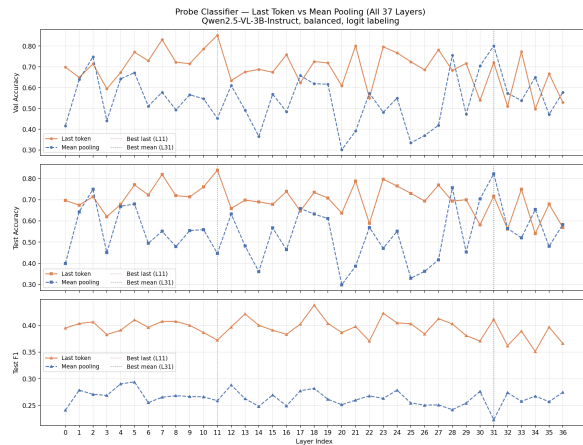


Figure 1: Layer ablation on Qwen2.5-VL-3B-Instruct. Last-token probing peaks at layer 11 with test accuracy 0.84.

Phi-3.5-Vision-Instruct. Phi has a distinct two-phase pattern: layers 0–8 fluctuate near chance, then both strategies stabilize sharply around layer

9. Last-token peaks at **layer 21** (val 0.891, test acc 0.878, test F1 0.693) while mean pooling peaks later at layer 31 (val 0.883, test acc 0.881, test F1 0.669) [Figure 2]. Unlike Qwen, the gap between the two strategies is small, suggesting that hallucination information in Phi is distributed more evenly across token positions rather than concentrated in the last token.

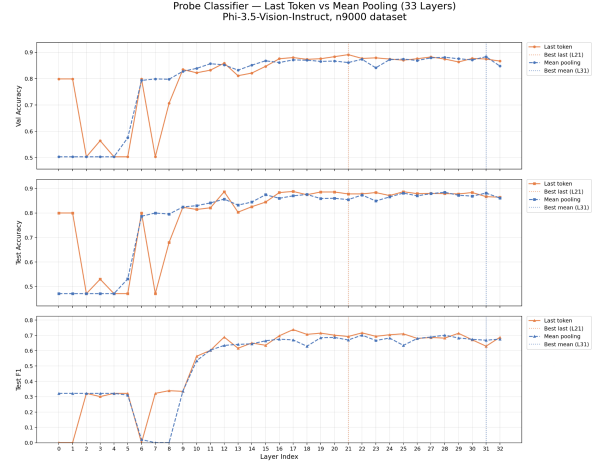


Figure 2: Layer ablation on Phi-3.5-Vision-Instruct. Both strategies are competitive; last-token peaks at layer 21.

InternVL2-2B. InternVL2-2B shows a late-emerging signal. The first two layers achieve high apparent accuracy but near-zero F1, meaning the probe predicts only the majority class. Accuracy drops toward chance between layers 2–9, then rises steadily, with both strategies peaking at the **final layer 24** (last-token: val 0.876, test acc 0.882, F1 0.634; mean pooling: val 0.879, test acc 0.883, F1 0.637). [Figure 3] The near-identical performance of both methods at this final layer suggests that by layer 24 the hallucination signal has been fully diffused across all token positions.

DeepSeek-VL-1.3B-Base. DeepSeek shows a clean monotonically increasing layer curve, peaking at **layer 23** (val 0.857, test acc 0.851, test F1 0.868) before dropping slightly at the final layer. [Figure 4] Because DeepSeek generates binary yes/no answers as a single output token, mean pooling over output tokens is mathematically identical to last-token, making the pooling comparison invalid for this model. The result should therefore be interpreted solely as a layer-ablation finding: despite being the smallest model (1.3B), DeepSeek yields a strong and clean late-layer hallucination

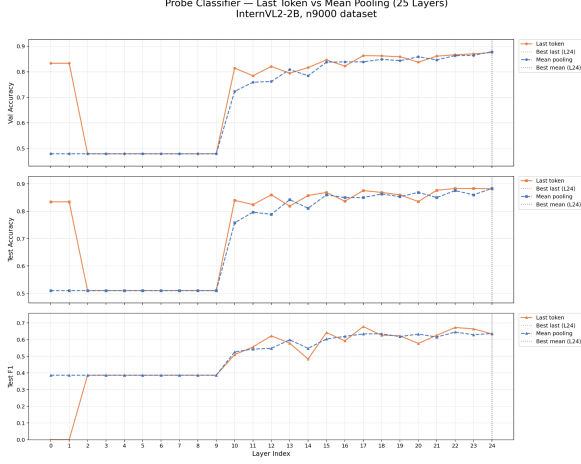


Figure 3: Layer ablation on InternVL2-2B. Both strategies peak at the final layer; signal emerges only after layer 9.

signal with the highest test F1 among all five models. Also, see the reasoning of the final layer accuracy drop in the MiniCPM section below.

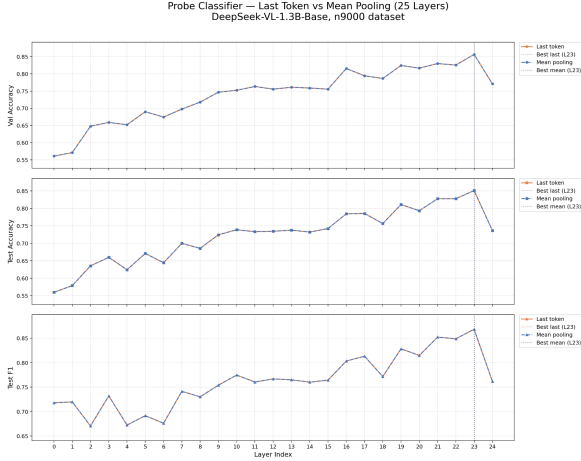


Figure 4: Layer ablation on DeepSeek-VL-1.3B-Base. Useful late-layer signal. (Both line overlapping due to only single token output. Hence last token and mean pool have the same result.)

MiniCPM-V2. MiniCPM was evaluated on a smaller balanced subset ($N = 434$) due to extraction instability on limited-compute hardware. Early layers are noisy with near-zero F1, but layers 28–39 become strong and stable. The best layer is **layer 32** (val 0.875, test acc 0.821, test F1 0.596) [Figure 5]. As with DeepSeek, the pooling comparison is invalid for this model for the same single-token-output reason. Results should be interpreted cautiously given the smaller sample size. Moreover, notice the signal strength drop in the last layer. We suspect that the final layer is optimized for gen-

erating the next answer token, not necessarily for preserving all grounding information. So The best hallucination signal appears slightly before the final layer, where the model still retains richer image-question reasoning features.

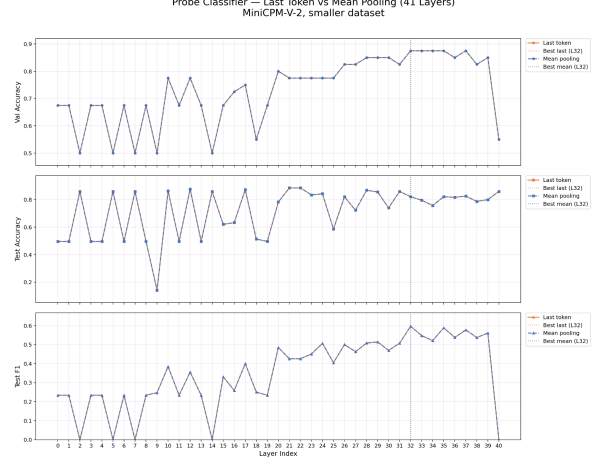


Figure 5: Layer ablation on MiniCPM-V2. Useful late-layer signal. (Both line overlapping due to only single token output. Hence last token and mean pool have the same result.)

5.1.2 Summary of Best-Layer Results

Table 1 summarizes the best probe performance across all five models.

Model	Best Layer	Val Acc	Test Acc	Test F1
Qwen2.5-VL-3B	11	0.850	0.840	0.370
Phi-3.5-Vision	21	0.891	0.878	0.693
InternVL2-2B	24	0.876	0.882	0.634
DeepSeek-VL-1.3B	23	0.857	0.851	0.868
MiniCPM-V2	32	0.875	0.821	0.596

Table 1: Best-layer probe results per model (last-token features, except DeepSeek and MiniCPM where pooling is not applicable). All models surpass the naïve majority-class baseline.

5.2 Discussion

5.2.1 Short-cut analysis

A central concern for any probing study is whether the classifier learns the intended signal or merely a shortcut. Our labels are balanced, so the classifier cannot achieve high accuracy by simply predicting a single class. However, balanced labels alone do not rule out shortcuts. The classifier may still find an easy correlation in the data that yields high accuracy without reflecting any true understanding of hallucination.

We identify two candidate shortcuts in the POPE setting. The first is to predict hallucination when-

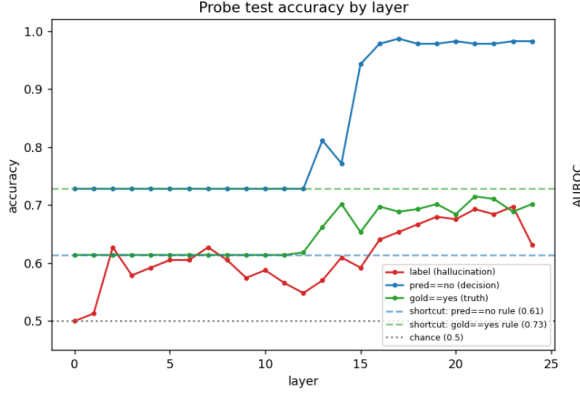


Figure 6: Shortcut analysis across layers.

Pred	Gold	Label	n
0	0	0	349
0	1	1	478
1	0	1	90
1	1	0	219

Table 2: Distribution of the model’s prediction (Pred), the ground-truth answer (Gold), and the derived hallucination label

ever the model answers no. On our test split this rule alone reaches roughly 61% accuracy. The second is to predict based on whether the ground-truth answer is yes, which reaches roughly 73%. If our classifier were silently relying on either pattern, its apparent performance would be misleading.

Table 2 reports the joint distribution of these quantities on the test split, from which both shortcut baselines can be understood directly. Always predicting “hallucination” when the model answers “no” corresponds to the rows where the prediction is negative, giving $(478 + 219)/(349 + 478 + 90 + 219) \approx 0.61$. Always predicting from whether the ground truth is “yes” gives $(478 + 349)/(349 + 478 + 90 + 219) \approx 0.73$. These are exactly the two dashed baselines in Figure 6.

Figure 6 compares three quantities across layers. The blue curve measures how well the classifier predicts the model’s own yes/no decision. It rises to nearly 98% in later layers, confirming that the pre-decoding hidden states already encode what the model is about to answer. The red curve is the quantity we actually care about, hallucination-detection accuracy, and the green curve shows how well the classifier can predict the ground-truth answer. The question is whether the red curve is simply tracking one of the two shortcuts.

The comparison is reassuring on both counts. First, the red curve exceeds the predict-on-no shortcut baseline of 61%, indicating that the classifier does more than copy the model’s own decision. Second, although the red curve sits below the 73% ground-truth-is-yes baseline, the classifier has no means of exploiting that rule: following it would require knowing the true answer, and the green curve shows the classifier can recover it only about 68% of the time, it cannot achieve 73% without correct ground-truth. Taken together, these results support that the classifier’s hallucination signal is genuine rather than an artifact of either shortcut.

5.3 Further Validation for the Probes

The previous results are based on POPE, which only asks the model to answer in yes or no and assumes hallucination whenever the model answers incorrectly. To better evaluate probe performance, we used MMHal-Bench (Sun et al., 2024), which includes 96 open-ended image QA questions and uses an LLM to assess the model’s answer against a given ground truth. The classifier for Qwen2.5-VL-3B using mean pooling and last-token features achieves 49.46% and 58.06% accuracy, respectively. While the classifier using last-token features still outperforms the one using mean-pooling, the accuracy is overall very low, indicating the probes are not effectively identifying hallucinations.

These results provide potential evidence that the classifier trained on POPE may not effectively generalize to general hallucination detection. Further studies could be done on interpreting the trained classifiers, such as the correlation between predicting hallucination and the model’s confidence in generating the labels.

5.3.1 Limitation

Dataset simplicity. We trained the probes using the POPE dataset, which frames hallucination detection as a binary yes/no question about whether an object is present. This formulation gives clean, deterministic labels and makes evaluation straightforward, but it is also coarse. Because the answer space is just “yes” or “no,” an incorrect response cannot be unambiguously attributed to hallucination: the model may be hallucinating an absent object, but it may equally have made an ordinary perceptual or reasoning error, or simply guessed wrong. Our binary labels therefore conflate genuine hallucination with other failure modes, which limits how precisely we can claim to be detect-

ing hallucination specifically. Given more time, we would construct a more complex free-response dataset, where the model produces open-ended descriptions and an external model (e.g., a stronger LLM accessed via API) judges whether each response contains hallucinated content. This would yield labels that are more faithful to hallucination as a phenomenon, at the cost of a more involved and less deterministic labeling pipeline.

Model scale. The original study (Alnuhait et al., 2025) uses language models up to 70B parameters, whereas our compute budget (Colab) restricted us to VLMs in the 1.3B–4.2B range. Smaller models may carry less informative hidden states, and indeed the layer at which hallucination signal emerges varied considerably across our models rather than following a single consistent trend. Our findings should therefore be read as evidence about small VLMs, and may not transfer directly to larger models.

Implementation gaps. Several training details are not fully specified in the original paper, leaving room for interpretation in our reimplementations. Where details were ambiguous, we made reasonable choices, but this means our setup is not guaranteed to match the original exactly, and some differences in results may stem from these implementation decisions rather than from the shift to the multimodal setting.

6 Conclusion

We extended the core finding of FACTCHECKMATE to the multimodal setting, demonstrating that frozen VLM hidden states encode a detectable hallucination signal before any output token is generated. Across five architecturally diverse VLMs ranging from 1.3B to 4.2B parameters, a simple two-layer MLP probe trained on class-balanced hidden states consistently achieves test accuracies between 0.82 and 0.88 on the POPE benchmark. The optimal probing layer varies by architecture: Qwen concentrates its signal in a mid-network layer (layer 11 of 37), Phi and DeepSeek build it gradually toward the later layers, and InternVL2 produces a useful signal only after deep multimodal fusion near the final layer. Last-token probing outperforms mean pooling for most models, though the gap narrows for architectures with more distributed representations.

A shortcut analysis confirms that the probe’s

accuracy cannot be explained by trivial decision rules based on the model’s own yes/no output or the gold answer distribution. These results suggest that preemptive hallucination detection, originally validated on text-only LLMs, transfers meaningfully to vision-language models despite the added complexity of multimodal token streams. Future work could investigate whether detector-triggered activation steering can reduce hallucination rates at inference time without retraining the base model.

7 Group Member Roles

- **Bochao Cai:** dataset preparation, POPE benchmark and evaluation.
- **Gezheng Kang:** classifier training, layer ablations, error analysis, generation scripts.
- **Luca Paliska:** dataset preparation, POPE benchmark and evaluation.
- **Samuel Huang:** classifier training, layer ablations, error analysis, generation scripts.
- **Qiyuan Tan:** classifier training, layer ablations, error analysis, generation scripts.

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